FPT School of Business and Technology

FINAL PROJECT

APPENDIX II - USER GUIDE FOR DEPLOYMENT AND OPERATION

Course	:	DDM501.22 - AI in DevOps, DataOps, MLOps
Project topic	:	Predict Customer Satisfaction / Trust Toward AI in Online Shopping
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Hanoi, March 2026

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1 OVERVIEW

This system is an end-to-end Machine Learning application designed to predict customer satisfaction toward AI in online shopping. The solution integrates data processing, model training, experiment tracking, API deployment, and monitoring.

The system leverages:

- MLflow for experiment tracking
- FastAPI for model serving
- Prometheus and Grafana for monitoring
- Docker Compose for multi-service deployment

2 PREREQUISITES

Before deploying the system, ensure the following requirements are met:

- Python 3.11 installed
- PowerShell (for Windows users)
- Docker Desktop (for containerized deployment)
- Kaggle API credentials configured (for automatic dataset download)

3 LOCAL ENVIRONMENT SETUP

Create virtual environment

```
python -m venv venv
.\venv\Scripts\activate
```

If the environment is not activated correctly:

```
.\venv\Scripts\python.exe main.py
```

Install dependencies

```
pip install --upgrade pip
pip install -r requirements.txt
```

4 DATASET SETUP

The system uses the Kaggle dataset:

- pooriamst/online-shopping

Option 1: Manual setup

Place the dataset manually:

```
data/raw/online_shopping_ai.csv
```

Option 2: Automatic download

Run training directly. The system will:

- Download dataset using kagglehub
- Copy CSV to the required location
- Continue ingestion and training

Optional configuration:

```
$env:KAGGLE_DATASET="pooriamst/online-shopping"
```

Troubleshooting (Kaggle error 403)

Possible causes:

- Missing or incorrect Kaggle credentials
- Dataset access not granted
- Incorrect dataset configuration

5 MODEL TRAINING AND EXPERIMENT TRACKING

Start MLflow server

```
mlflow server \
--host 127.0.0.1 \
--port 5000 \
--backend-store-uri sqlite:///mlflow.db \
--default-artifact-root ./mlruns
```

Tracking URI:

```
MLFLOW_TRACKING_URI=http://localhost:5000
```

Run training pipeline

```
python pipeline/train.py
```

Training pipeline includes:

- Data ingestion
- Schema normalization
- Train/validation/test split
- Model training:
    - Logistic Regression

  - Random Forest
  - LightGBM
- Model evaluation
- Experiment logging to MLflow

Output artifacts

```
artifacts/
├── model.joblib
├── preprocessor.joblib
├── label_encoder.joblib
```

6 MODEL SERVING (API OPERATION)

Start API service

```
python main.py
```

Available endpoints

- `GET /api/v1/health` → system health check
- `GET /api/v1/model/info` → model metadata
- `POST /api/v1/predict` → single prediction
- `POST /api/v1/predict/batch` → batch prediction
- `GET /metrics` → Prometheus metrics

Example request

```
{
  "country": "INDIA",
  "online_consumer": "YES",
  "age_group": "Gen X",
  "annual_salary_band": "Medium High",
  "gender": "Female",
  "education": "Masters' Degree",
  "payment_method_card": "YES",
  "living_region": "Metropolitan",
  "online_service_preference": "YES",
  "ai_endorsement": "YES",
  "ai_privacy_no_trust": "NO",
  "ai_enhance_experience": "YES",
  "ai_tool_chatbots": "YES",
  "ai_tool_virtual_assistant": "YES",
  "ai_tool_voice_photo_search": "NO",
  "payment_method_cod": "NO",
  "payment_method_ewallet": "YES",
  "product_category_appliances": "NO",
  "product_category_electronics": "YES",
  "product_category_groceries": "NO",
  "product_category_personal_care": "YES",
  "product_category_clothing": "NO"
}
```

Local access URLs

- API Docs: http://localhost:8000/docs
- Health Check: http://localhost:8000/api/v1/health
- Metrics: http://localhost:8000/metrics

7 TESTING

Run the test suite:

```
python -m pytest tests/test_system.py -q
```

Test coverage includes:

- Feature processing
- Model prediction
- API endpoints
- System integration

8 DEPLOYMENT WITH DOCKER COMPOSE

Recommended workflow

1. Train the model locally
2. Verify artifacts exist
3. Deploy system

Start system

```
docker compose up -d --build
```

Services

- API → http://localhost:8000
- MLflow → http://localhost:5000
- Prometheus → http://localhost:9090
- Grafana → http://localhost:3000

Stop system

```
docker compose down
```

9 MONITORING AND OPERATION

The system integrates monitoring tools:

- Prometheus collects:
  - Request count
  - Latency
  - Error rate
- Grafana provides:
  - Dashboards
  - Performance visualization
  - Real-time monitoring